जम्मू केंद्रीय विश्वविद्यालय

Central University of Jammu

रह्या-सुचानी (बगला), जिला सांबा -181143, (जम्मू और कश्मीर)

Rahya-Suchani (Bagla), District: Samba – 181143, Jammu (J&K)

14 July 2025

Roll No. | | | | | | | | | | | | |

SEMESTER MAJOR EXAMINATIONS

Programme	:	B. Tech. (Computer Science & Engineering)	Semester	:	VI
Course Name	:	Data Mining and warehousing	Course Code	:	BECSE0O011
Duration	:	3 Hours	Credits	:	04
Maximum Marks	:	100	Minimum Marks	:	-

After pursuing this course, the student will be able to:	
CO 01	Understand data warehousing concepts, architecture, and implementation.
CO 02	Design and manage data warehouses to support efficient data analysis and retrieval.
CO 03	Apply data mining techniques to discover patterns, clusters, and associations.
CO 04	Evaluate and improve data mining models for better accuracy and knowledge extraction.
CO 05	Solve real-world business problems using data mining and warehousing methods.

Part A Very Short Answer Type Logical Questions of 04 marks related to different COs		Total Marks 4X5=20	
S.No.	Statement	Levels	CO
1.	Differentiate between operational database and data warehouse.	L1	01
2.	Explain ROLAP, MOLAP and HOLAP server architecture.	L1	01, 02
3.	Consider a FFNN having one hidden layer with 3 neurons, Sigmoid function as the activation function and cross entropy as the loss function and input layers comprises of the two nodes, X1 and X2 where X1 = 3.8 and X2 = 2.9 and target class label, y =1. Initialize weights and bias between -1 and 1. Compute the predicted class label and loss at output layer.	L3	03
4.	Explain the role of principal components analysis in dimensionality reduction.	L2	01, 02
5.	Discuss the Market Basket Analysis with suitable example.	L1	03
PART B Short Answer Type Analytical Questions of 8 marks related to different COs		Total Marks 8X5=40	
6.	What is data cube? Explain various data cube operations with suitable examples.	L2	01
7.	Draw and explain three tier architecture of Data warehouse.	L1	01, 02,
8.	a.) How does a Support Vector Machine (SVM) classify data? (4) b.) Discuss the importance of kernel functions in Support Vector Machines. (4)	L1, L2	03
9.	Explain hashing and transaction reduction technique for improving the efficiency of Apriori algorithm for transaction database (given in Part C Q no.11. c.).	L4	04

10.	a.) Draw and explain Star, Snowflake and fact constellation Schema with suitable example (5) b.) Apply Haar-2 wavelet transformation to compute approximation (A) and detail (D) coefficient of the signal x= [4,6,10,12,14,16,18,20]. (3)	L1, L3	01, 03
	Suppose that a group of 1500 people were surveyed. The gender of each person was noted. Each person was polled as to whether his or her preferred type of movie was Action or Thriller. Thus, we have two attributes, gender and preferred movie. The observed frequency (or count) of each possible joint event is summarized in the contingency table below. Apply the chi-square test to check weather two attributes are correlated or not. (Critical value at 0.001 level of significance and 1 degree of freedom is 10.828)	L4	03, 05

	Male	Female
Action	250	200
Thriller	50	1000

Part C

Long Answer Type Descriptive Questions of 20 marks related to different Cos

(Only Two questions to be attempted)

Total Marks 20X2= 40

11.	a.) Write and explain the pseudocode of Apriori algorithm. (5) b.) What are ensemble methods? Discuss the advantages of ensemble learning in classification. (5) c.) Apply the Apriori algorithm to generate the frequent itemset with minsup=3 and strong association rule with minconf=50% from the following transaction database. (10)	L1, L2, L5	03,04 ,05

Trans Id.	Items Purchased
T1	{I1, I2, I5}
T2	{I1, I2, I4}
T3	{I1, I2}
T4	{I2, I5}
T5	{I1, I3, I4}
T6	{I2, I3}
T7	{I1, I2, I3, I5}
T8	{I3, I5}

12.	a.) Write and explain the basic algorithm for inducing a decision tree. (5) b.) Explain the significance of Gini index and Gini ratio in decision tree splitting. (5) c.) Given the following dataset construct the decision tree using information gain: (10)	L1, L3, L5	03,04 ,04

Weather	Temperature	Humidity	Wind	Play Tennis?
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rainy	Mild	High	Weak	Yes
Rainy	Cool	Normal	Weak	Yes
Rainy	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	Normal	Weak	No
Sunny	Cool	High	Weak	Yes
Rainy	Mild	Normal	Weak	Yes

13. a.) Define hierarchical clustering and differentiate between agglomerative and divisive clustering methods. (5)

b.) What is outlier? Explain how Partitioning Around Medoids (PAM) clustering algorithm can be used for outlier detection. (5)

c.) Apply the naïve Bayesian classifier to classify a tuple X= (age=youth, income=medium, student=yes, credit rating=fair) from the following data (10)

RID	age	income	student	credit_rating	Class: buys_computer
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes

L1, 12, L5

03,04 ,05